

Homework 5 Math 210A

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Problem 1. Let $\varphi : G_1 \rightarrow G_2$ be a group homomorphism.

(a) Show that if $N_2 \trianglelefteq G_2$, then $\varphi^{-1}(N_2) \trianglelefteq G_1$. Deduce that $\ker(\varphi) \trianglelefteq G_1$.

Proof. Suppose $N_2 \trianglelefteq G_2$. Our goal is to show that $\varphi^{-1}(N_2) \trianglelefteq G_1$; that is, for all $g \in G_1$,

$$g\varphi^{-1}(N_2)g^{-1} \subseteq \varphi^{-1}(N_2). \quad (*)$$

To this end, let $g \in G_1$ be arbitrary. To show (*), let $x \in g\varphi^{-1}(N_2)g^{-1}$ be arbitrary. Then

$$x = g\tilde{x}g^{-1} \text{ for some } \tilde{x} \in \varphi^{-1}(N_2).$$

Since $\tilde{x} \in \varphi^{-1}(N_2)$, it follows that $\varphi(\tilde{x}) \in N_2$. But note that $N_2 \trianglelefteq G_2$ and so for all $g' \in G_2$, we have

$$g'N_2(g')^{-1} = N_2. \quad (\dagger)$$

Since φ is a homomorphism, we see that

$$\varphi(x) = \varphi(g\tilde{x}g^{-1}) = \varphi(g)\varphi(\tilde{x})(\varphi(g))^{-1}$$

Since $\varphi(g) \in N_2$, we have $\varphi(x) \in N_2$ from $(\dagger)$. Thus, $x \in \varphi^{-1}(N_2)$ and so we conclude that

$$g\varphi^{-1}(N_2)g^{-1} \subseteq \varphi^{-1}(N_2)$$

which is our desired result. We see that $\ker(\varphi) \trianglelefteq G_1$ because

$$\begin{aligned} \ker(\varphi) &= \{g \in G_1 : \varphi(g) = 1_{G_2}\} \\ &= \{g \in G_1 : g\varphi^{-1}(N_2)\} \\ &= \{g \in G_1 : g \in \varphi^{-1}(N_2)\} \\ &= \varphi^{-1}(N_2) \\ &\trianglelefteq G_1. \end{aligned}$$

■

(b) Show that if $N_1 \trianglelefteq G_1$, then $\varphi(N_1) \trianglelefteq \varphi(G_1)$. Deduce that any surjective homomorphism will map normal subgroups in the domain to normal subgroups in the codomain.

Proof. Suppose that $N_1 \trianglelefteq G_1$. To show that $\varphi(N_1) \trianglelefteq \varphi(G_1)$, we need to show that for all $g' \in \varphi(G_1)$,

$$g'\varphi(N_1)(g')^{-1} \subseteq \varphi(N_1). \quad (*)$$

Let $g' \in \varphi(G_1)$. To show (*), let $y \in g'\varphi(N_1)(g')^{-1}$. Then for some $\tilde{y} \in \varphi(N_1)$, we have

$$y = g'\tilde{y}(g')^{-1}.$$

Since $\tilde{y} \in \varphi(N_1)$, then for some $\tilde{x} \in N_1$, we have $\tilde{y} = \varphi(\tilde{x})$. Furthermore, $g' \in \varphi(G_1)$ implies that there exists some $\tilde{g} \in G_1$ such that $g' = \varphi(\tilde{g})$. Since φ is a homomorphism, it follows that

$$\begin{aligned} y &= g'\tilde{y}(g')^{-1} = \varphi(\tilde{g})\varphi(\tilde{x})\varphi(\tilde{g})^{-1} \\ &= \varphi(\tilde{g})\varphi(\tilde{x})\varphi(\tilde{g}^{-1}) \\ &= \varphi(\tilde{g}\tilde{x}\tilde{g}^{-1}). \end{aligned}$$

Notice that $\tilde{g}\tilde{x}\tilde{g}^{-1} \in \tilde{g}N_1\tilde{g}^{-1}$. But since $N_1 \trianglelefteq G_1$, we have for all $g \in G$, $gN_1g^{-1} = N_1$. In particular, $\tilde{g}N_1\tilde{g}^{-1} = N_1$. Hence, $\hat{x} = \tilde{g}\tilde{x}\tilde{g}^{-1} \in N_1$. Hence, we can see that $y \in \varphi(N_1)$ and so we conclude that

$$g'\varphi(N_1)(g')^{-1} \subseteq \varphi(N_1).$$

Hence, we conclude that $\varphi(N_1) \trianglelefteq \varphi(G_1)$. ■

Problem 2. Let G be a group with $N \trianglelefteq G$.

- (a) Suppose $x \in G$ with $|x| = n < \infty$. Show that n divides the order of xN in G/N .

Proof. ■

- (b) Further show that the order xN in G/N is the smallest positive integer k such that $x^k \in N$.

Proof. ■

- (c) Show that if $G = \langle x \rangle$, then G/N is cyclic.

Proof. ■

Problem 3. Let G with center $Z(G)$. Recall that $Z(G) \trianglelefteq G$.

- (a) Prove that if $G/Z(G)$ is cyclic, then G is abelian.

Proof. ■

- (b) Show that if $|G| = pq$ where p and q are primes (not necessarily distinct) then either G is abelian or $Z(G) = \{1_G\}$.

Problem 4 (Section 3.2 Problem 5). Let H be a subgroup of G and fix some element $g \in G$.

- (a) Prove that gHg^{-1} is a subgroup of G of the same order as H .

Proof. ■

- (b) Deduce that if $n \in \mathbb{Z}^+$ and H is the unique subgroup of order n then $H \trianglelefteq G$.

Proof. ■

Problem 5 (Section 3.2 Problem 8). Prove that if H and K are finite subgroups of G whose orders are relatively prime then $H \cap K = 1$.

Proof. ■